Project Name: TapIn

Team Members:

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Problem (The "Why")

- **Current Issue:** Limited attendance machines located at fixed positions that create a single point of attendance registration. During peak hours, employees may need to queue, and the hardware requires regular maintenance.
- **Impact:** Alternative registering attendance method, locations are flexible but remain within the Formulatrix building area

Proposed Solution

- **Our Approach:** Develop a web-based Employee Attendance System that allows employees to register attendance by tapping their RFID/NFC employee card on an authorized NFC reader or NFC-enabled device.
- **Key Value:** Flexible attendance locations within the company building. Reuses existing employee RFID/NFC cards. Reduces dependency on dedicated fingerprint machines.

Core Features

- **Feature 1:** RFID/NFC Attendance Check-in. Employees register attendance by tapping their employee card on an NFC-enabled device or authorized RFID reader.
- **Feature 2:** Attendance History. Employees can view: Daily attendance, Monthly attendance, Check-in/check-out times, Attendance status, and Late arrivals. Administrators can search and filter attendance records by employee and date.

- **Feature 3:** The system performs automatic validation before accepting attendance. Validation includes:
    - Registered employee card
    - Authorized office network
    - Working hours
    - Duplicate attendance detection.
    - Invalid attendance requests are rejected or flagged for administrator review.

High-Level Architecture & Quality Assurance

- **Architecture Pattern:** Clean Architecture strictly adhering to SOLID principles.
- **Design Patterns Used:** Repository Pattern, Dependency Injection
- **Code Quality Goal:** Must Have 0 warnings as possible (scanned via static analysis), and 100 % Unit Test Coverage

Identified Challenges & Risks

- **Risk 1:** Different employee cards may use different RFID/NFC technologies, requiring compatibility testing with available readers and devices.
- **Risk 2:** Browser support for NFC varies across operating systems and browsers.
- **Risk 3:** Since the application is web-based, network outages may temporarily prevent attendance submission.